

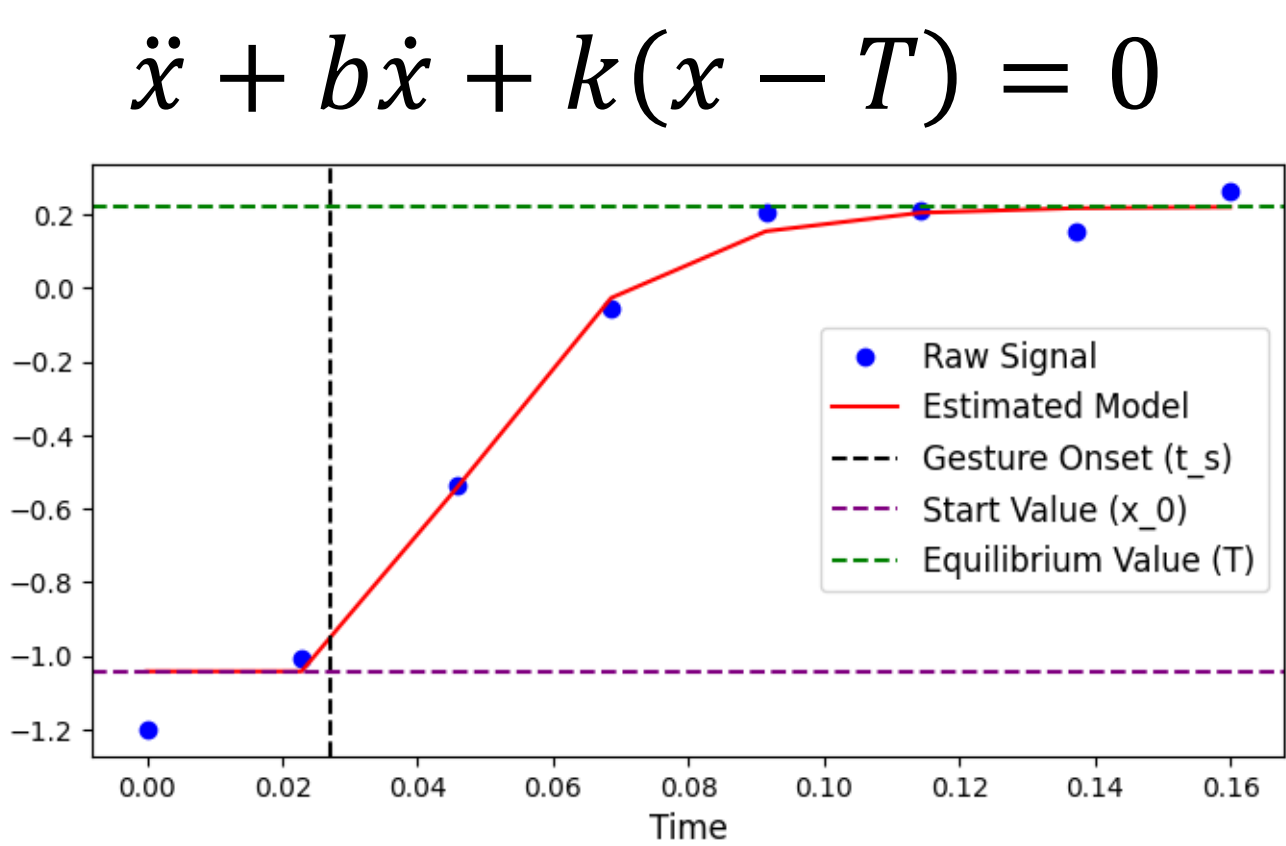
Applications and Challenges in Modeling Articulatory Dynamics

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Introduction

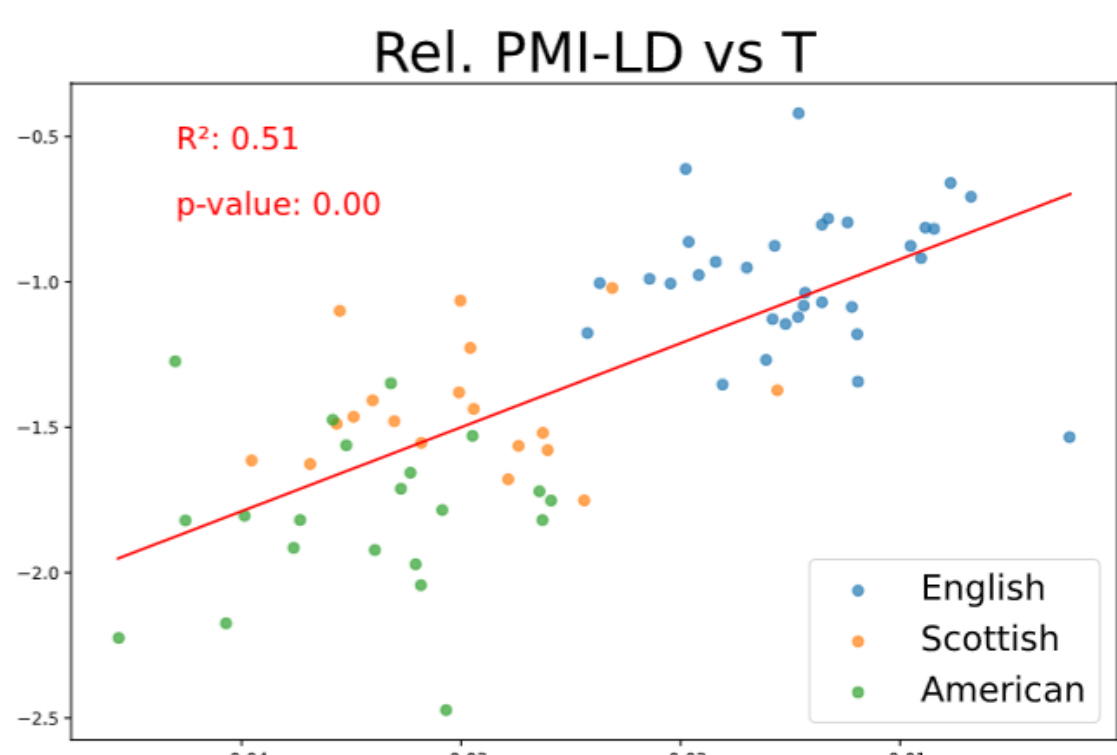
- Speech articulation has been modeled as a dynamical system (Saltzman & Munhall 1989; Browman & Goldstein, 1992; Iskarous, 2017), often in the form of a simple 2nd order spring-mass-damper



- Dynamical models offer an **interpretable** and **sparse** representation of articulatory signals, but desired parameters prove difficult to estimate outside of controlled setting

Previous Work

- **Goal:** Find systematic relationship between accent strength and articulatory features
- **Challenges:** Scarce articulatory data of speakers with varying levels of accent strength
- **Approach:**
 - Use articulatory inversion to approximate articulation from a dataset contain a range of speakers within two accents (VCTK)
 - Simplify analysis to vowel segments and calculate equilibrium value T under the dynamics model
 - Perform simple linear regression against PER (proxy for accent)



	TTA	TBA	TDA	TTR	TBR	TDR	LA, LP, JAW
/i/	T	-	↓	↓	-	↓	-
/v/	T	↑*	↑*	↑*	-	↓*	-
/ε/	T	↑	-	-	-	-	-
/ə/	T	-	-	-	-	-	-
/æ/	T	-	-	-	-	-	-
/ɪ/	T	-	-	-	-	-	↑LA*
/ɑ/	T	↓	↓	↓	-	-	-
/u/	T	-	-	-	-	↓*	↓JAW
/ʊ/	T	-	-	-	-	↓*	↓LP*
/ɜ/	T	↑*	↑*	↑*	↓*	↓*	↓LP

British /v/ has advanced tongue, "inconsistent" with retracted tongue implied by F2 studies

British /ɜ/ has advanced and lowered tongue, which aligns with expectations

American /ʊ/ has raised tongue dorsum and more lip protrusion, consistent with known lip rounding

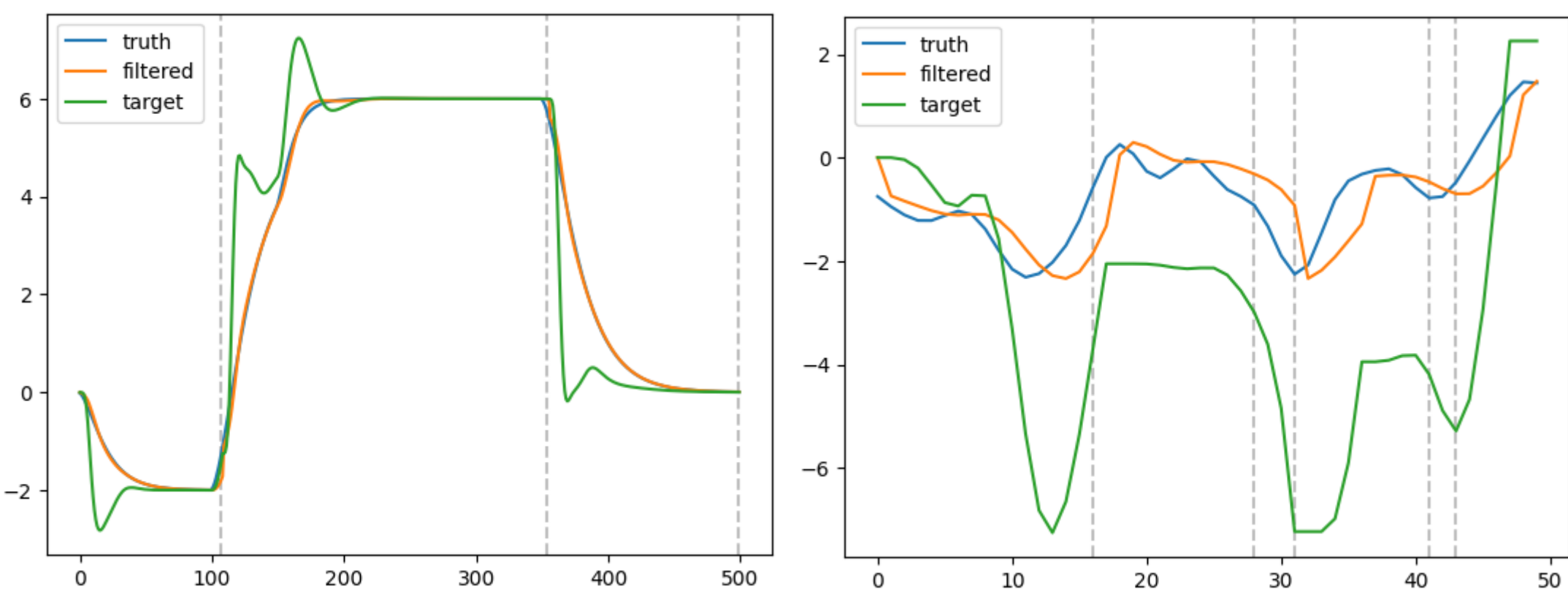
Ongoing Efforts

Kalman Filtering for Robust Parameter Estimation

- Classical spring-mass-damper formulation in Kalman filtering

$$\begin{aligned} s_{t+1} &= \begin{bmatrix} x_{t+1} \\ \dot{x}_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & \Delta t \\ -k\Delta t & 1 - b\Delta t \end{bmatrix} s_t + \begin{bmatrix} 0 \\ k\Delta t \end{bmatrix} T + w_t \\ y_t &= [1 \quad 0] s_t + v_t \end{aligned}$$

- Linearize desired parameters (b, k, T) and apply extended KF for parameter estimation
- System segmentation via change point detection (compare residual or NIS against moving threshold)



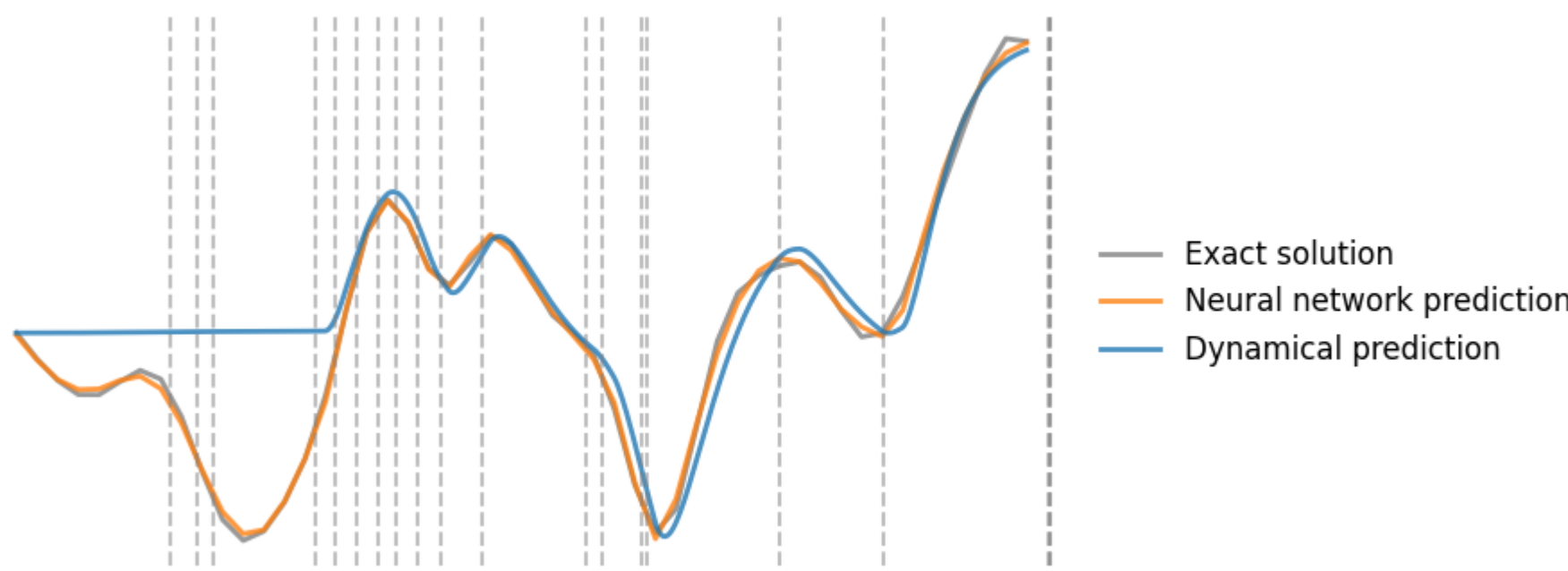
- Difficulties with fast movements at low sample rates (~50 Hz)

Physics-Informed Neural Networks for Holistic Estimation

- Predict signal from time input with a physics loss term $L_{physics}$ derived from the residuals of the differential equation

$$\begin{aligned} L &= L_{recon} + L_{physics} + L_{boundary} \\ L_{physics} &= \frac{1}{N} \sum_{i=1}^N \left| \frac{d}{dt^2} x(t_i, \theta) + b \frac{d}{dt} (t_i, \theta) + k(x(t_i, \theta) - T) \right|^2 \end{aligned}$$

- Learn parameters as a differentiable step function



- Difficulty obtaining consistent performance while being computationally expensive

Next steps

- Explore alternative methods for parameter estimation
 - Simple temporal segmentation via velocity estimation and iterative split & merge
 - Interacting multiple models as a Bayesian approach to system segmentation
- Potential steps towards application
 - Controllable speech editing/synthesis in the articulatory space via dynamics
 - Further analysis towards analyzing inter vs. intra speaker variability